

Feedback control

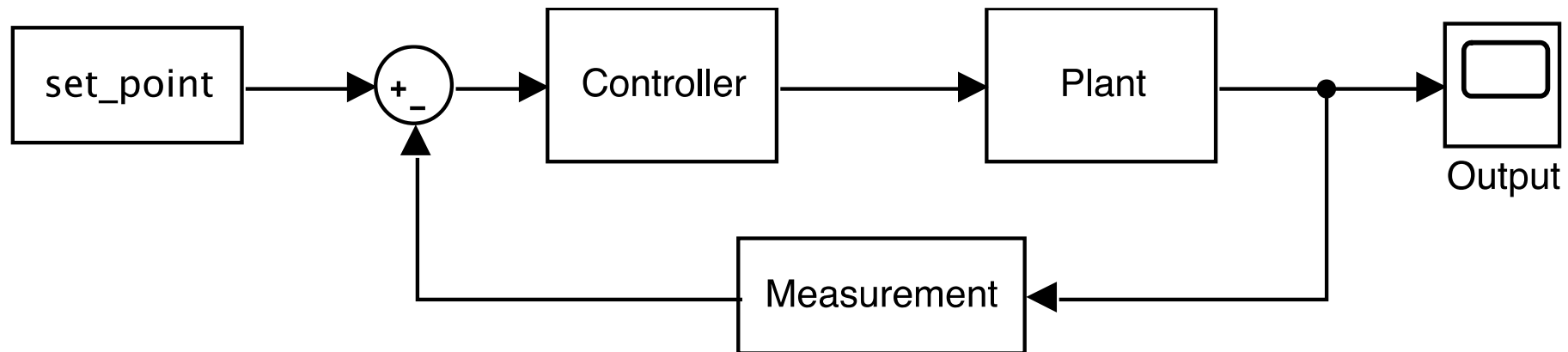
MEGN 441

Hot or cold?



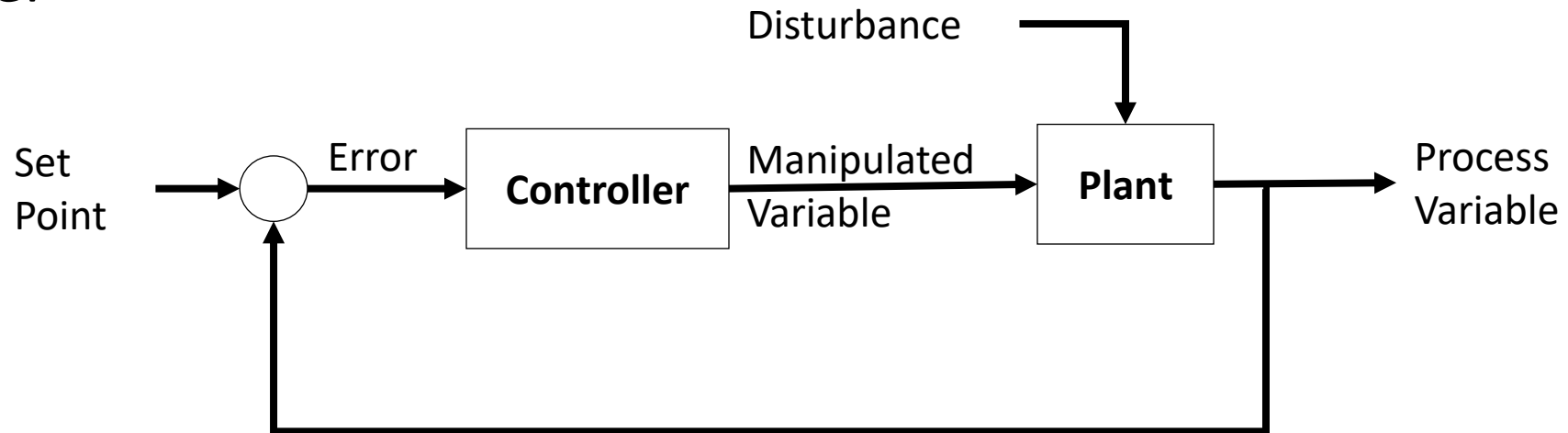
Feedback (closed-loop) control

- **Reads measurements** from the system indicating the current state
- Adjusts the control output to manipulate the system output to match the desired value
- Typically accomplished using various forms of weights on the error



Terminology

- **Process Variable:** the output of the plant
- **Set Point:** desired value for the process variable to achieve
- **Manipulated Variable:** the output of the controller, input of the plant
- **Disturbances:** inputs to the plant that change independent of the controller



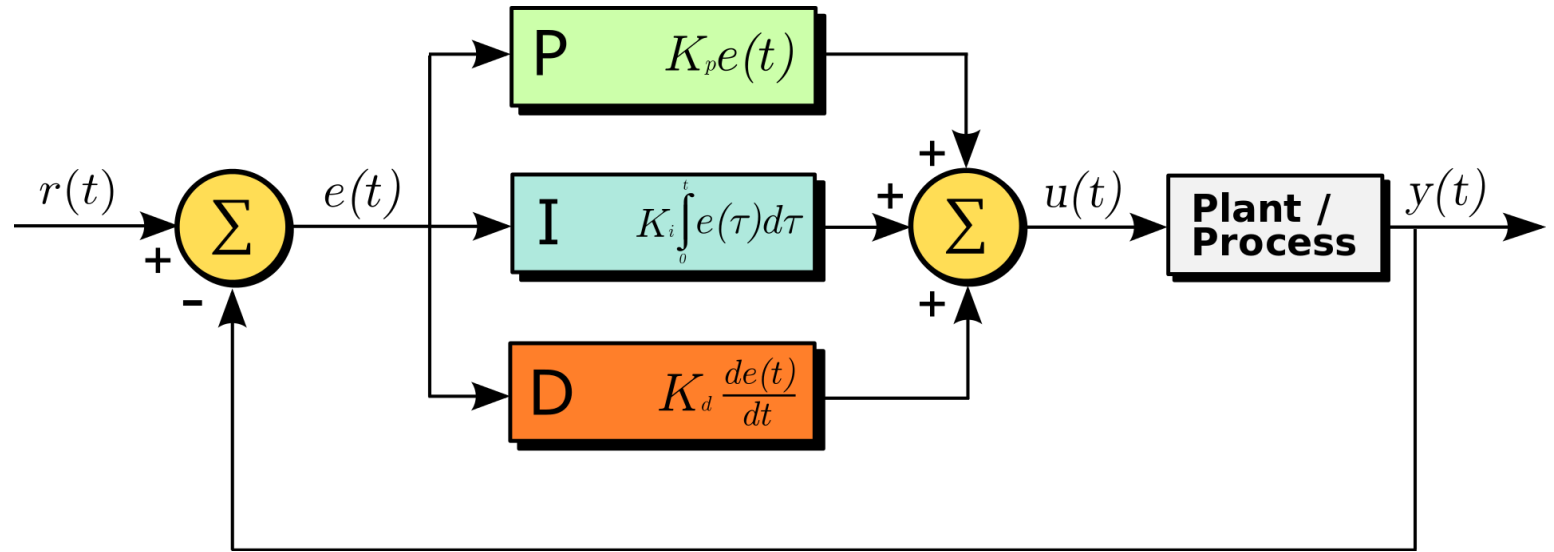
PID Controller

P: Proportional

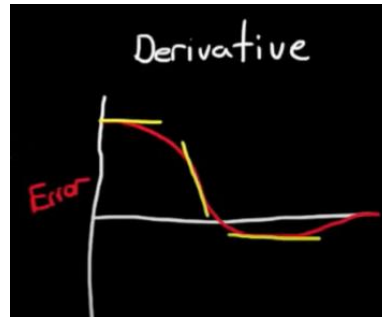
I: Integral

D: Derivative

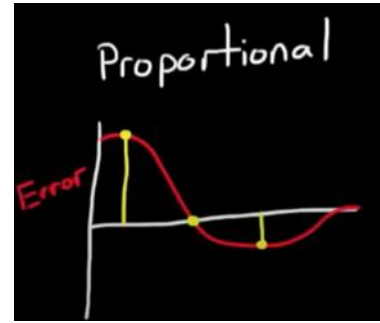
Very common controller
used in a variety of
situations



$$u(t) = \underbrace{K_p e(t)}_P + \underbrace{K_i \int_0^t e(\tau) d\tau}_I + \underbrace{K_d \frac{de(t)}{dt}}_D$$



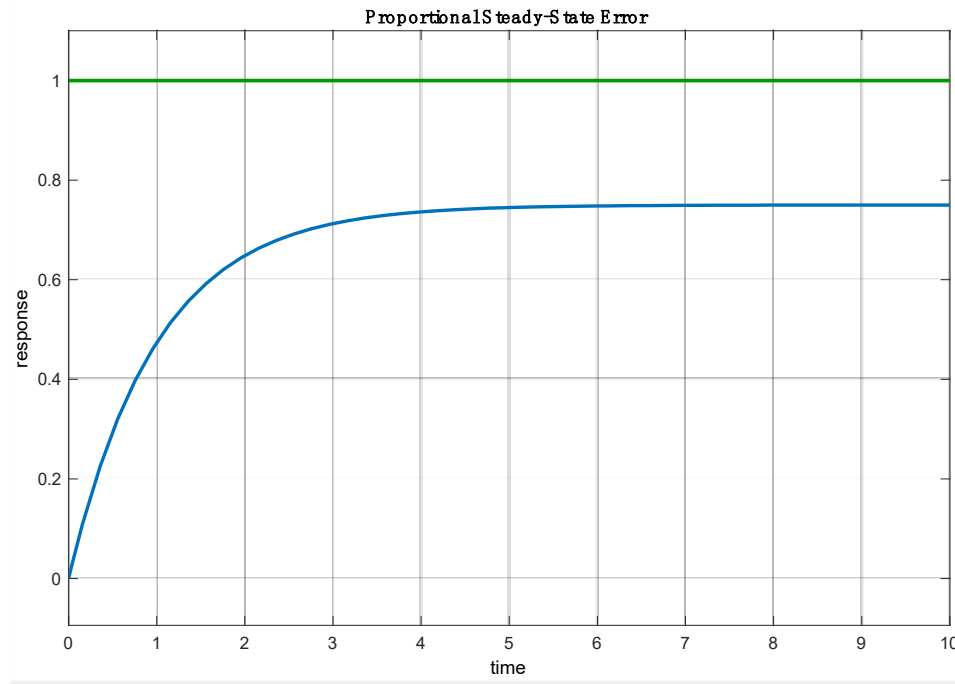
Proportional Term



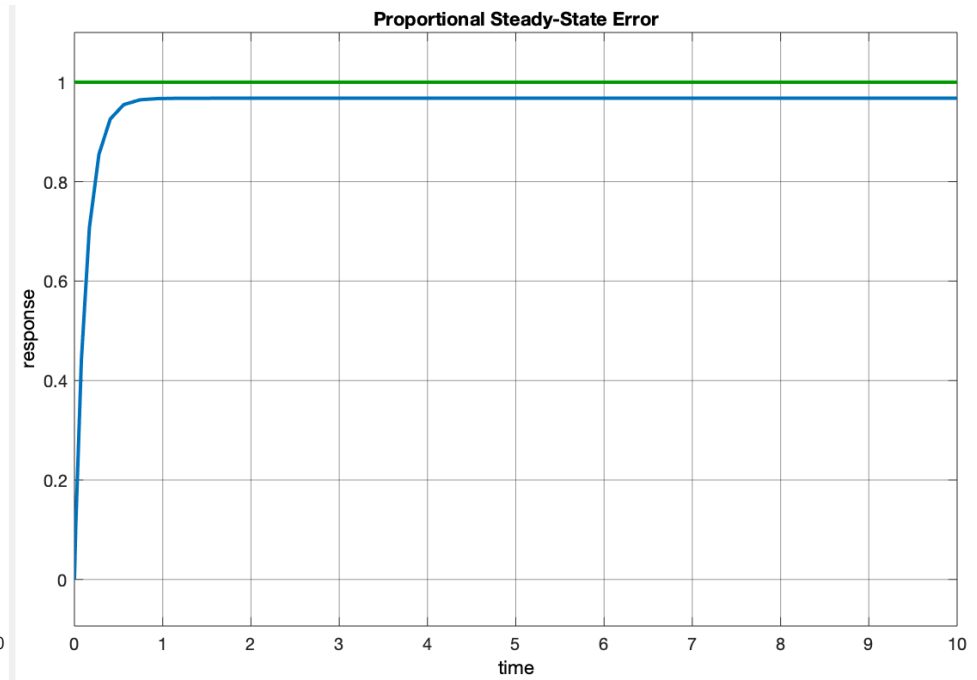
$$K_p e(t)$$

- Reacts to the **present error** $e(t)$.
- Changes the plant input based on how far off the process variable is from the set point.
- Because it requires some non-zero error to produce a value, this often leads to a steady-state offset error.
- A high enough gain may produce overshoot
- If plant response is quick, can lead to chatter around the set point
 - Remedied by using a “deadband”
 - Stop changing manipulated variable if within +/-d of set point

Steady-state error

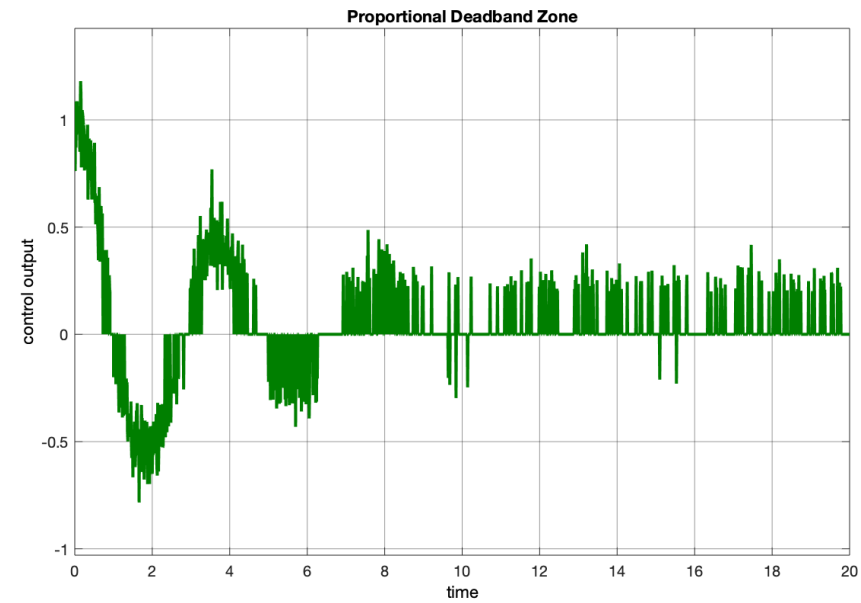
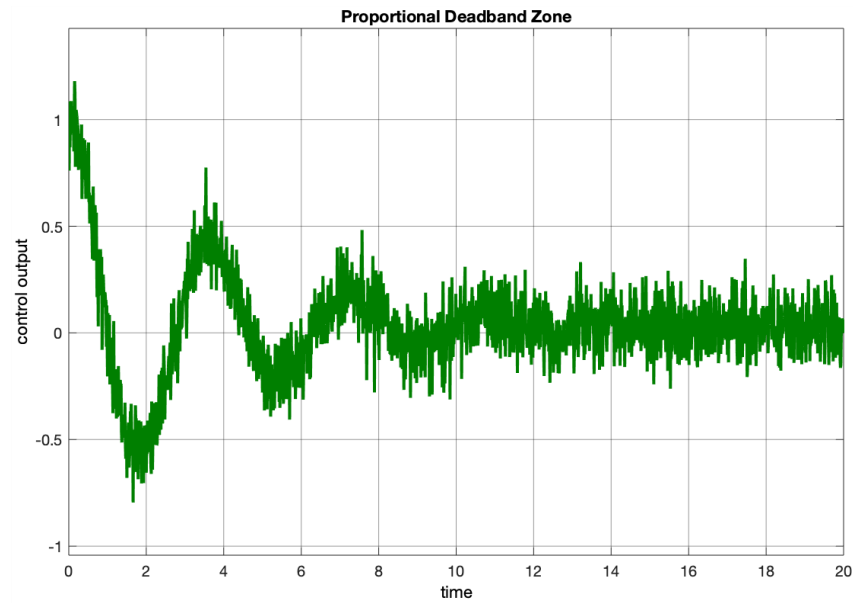
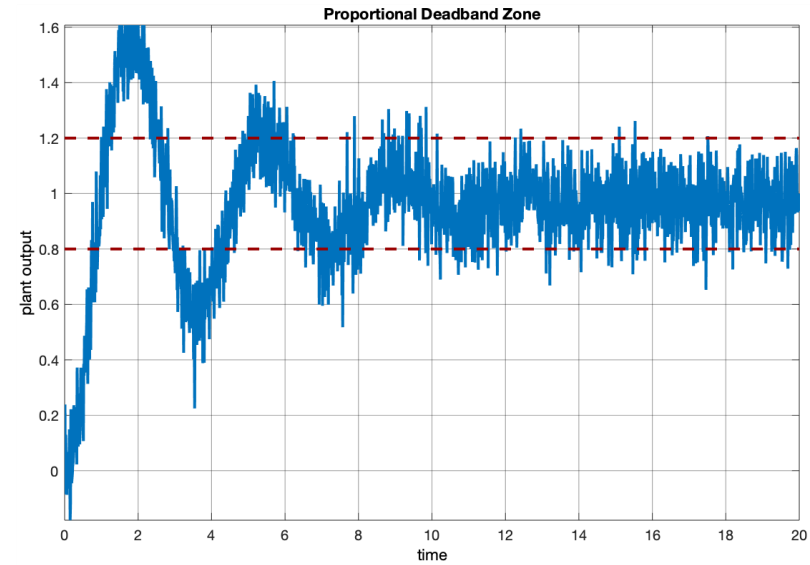


$$K_p = 1$$

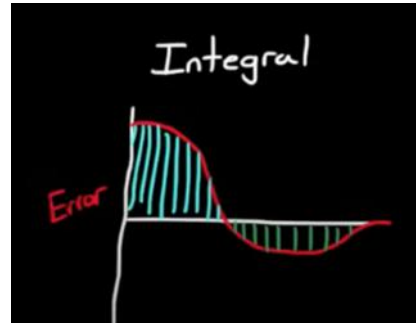


$$K_p = 10$$

Deadband (reduce chatter)



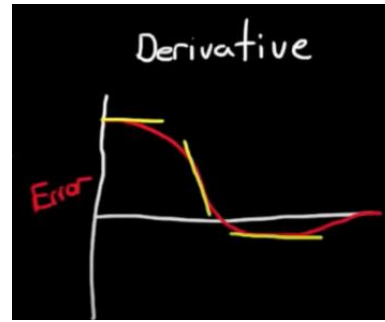
Integral Term



$$K_i \int_0^t e(\tau) d\tau$$

- Reacts to the sum of past errors (the area under the error curve).
- Changes the plant input based on the integral of the error
- Fixes steady-state offset
- Continues growing as long as the sign on the error stays constant

Derivative Term



$$K_d \frac{de(t)}{dt}$$

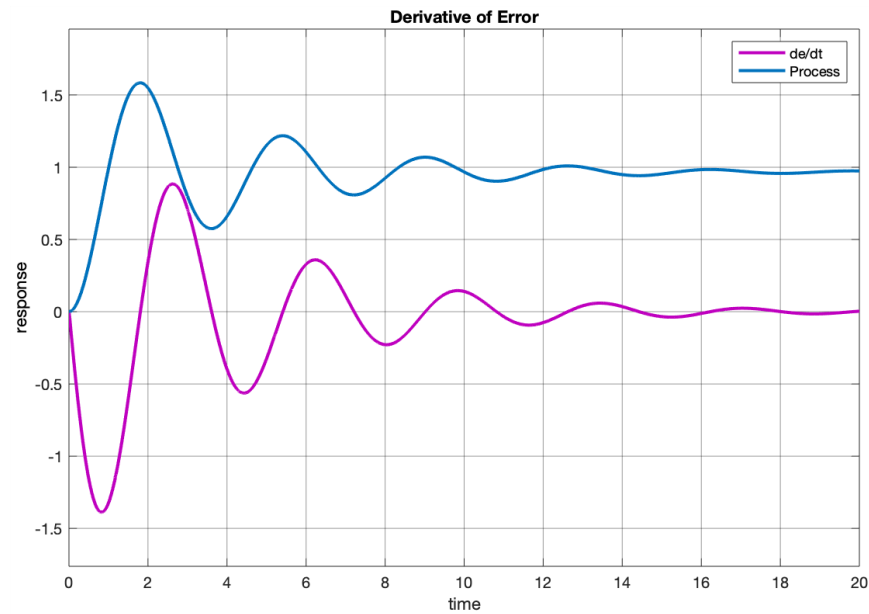
- Reacts to the rate of change of the error. Predicted Error Trend
- Changes the plant input based on the derivative of the error
- Helps to slow down the approach to the set point
- Can be susceptible to noise in the process variable
 - Consider adding a low-pass filter on the process variable
 - Hardware: RC circuit
 - Software: Exponentially Weighted Moving Average (EWMA)

<https://www.youtube.com/watch?v=4Y7zG48uHRo&t=79s>

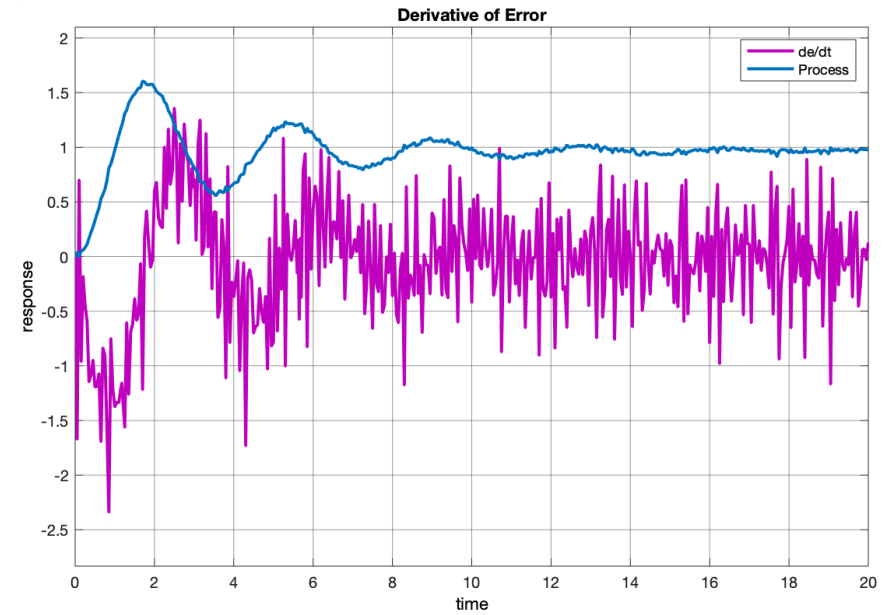
In summary

- P $\rightarrow$ Handles present error (fast response).
- I $\rightarrow$ Handles past error (eliminates steady offset).
- D $\rightarrow$ Handles future error trend (predictive, stabilizing).

Noise effects

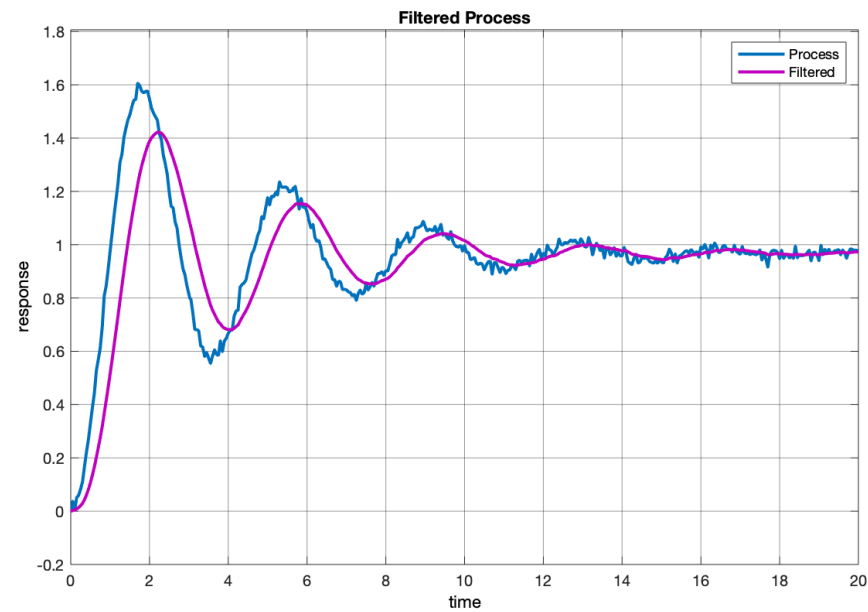


Plant + Derivative

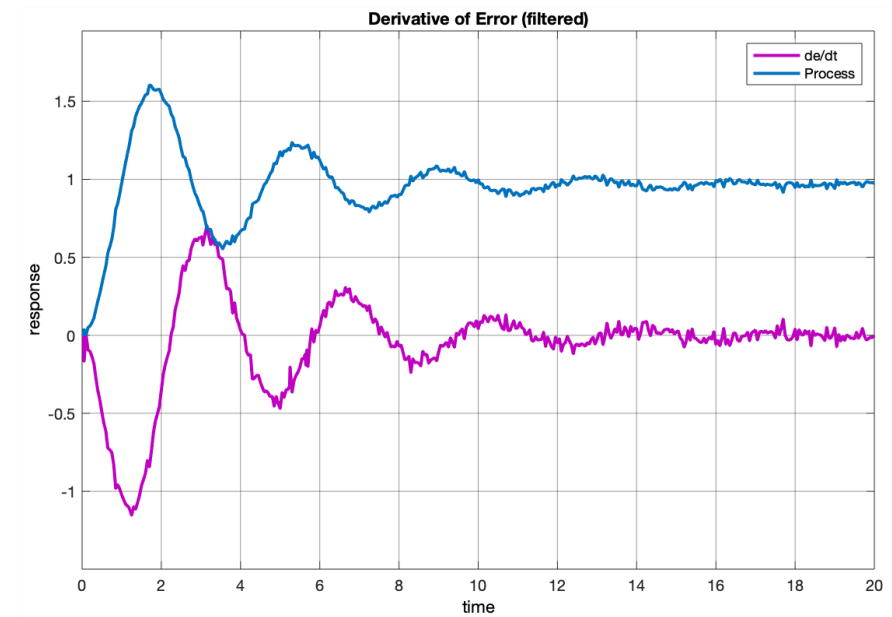


Derivative term from noisy plant

Noise effects



Noisy Plant + Filtered Plant



Derivative term from filtered plant

Other combinations

- PI Controller ($K_d = 0$)
 - Chemical Processes
 - Reduce steady-state errors, may reduce stability
- PD Controller ($K_i = 0$)
 - Robot arm movement, trajectory tracking
 - Integral less useful when set point is constantly changing

Proportional

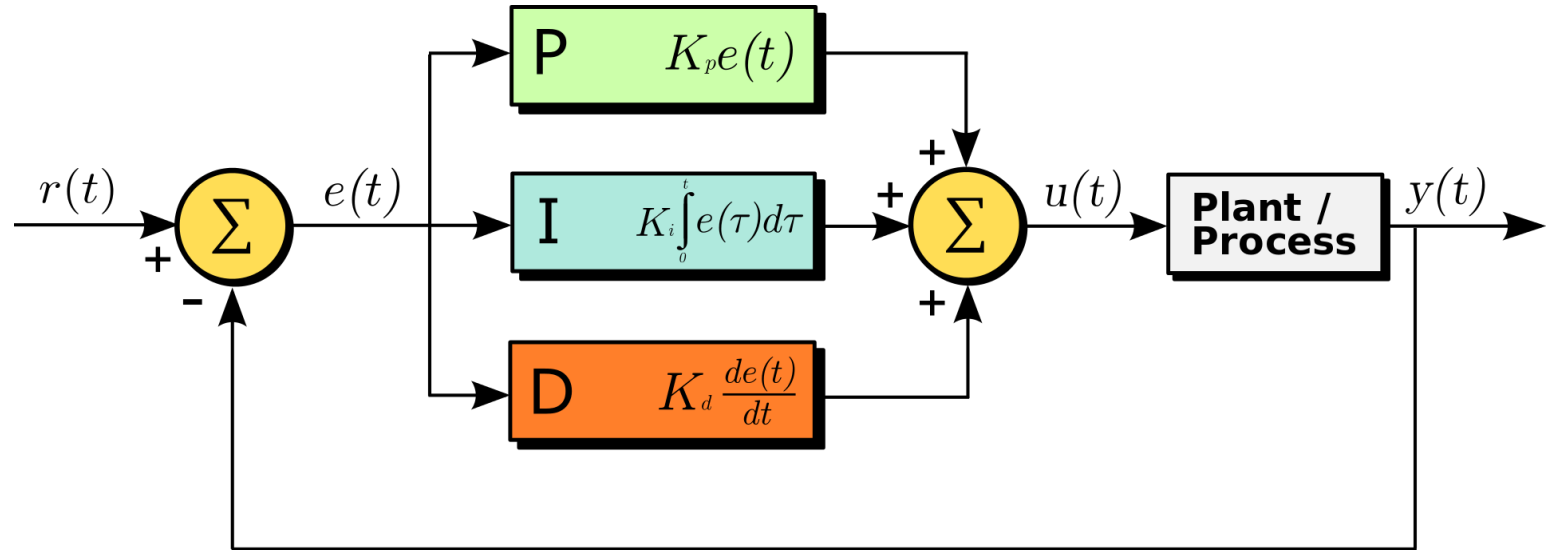
- Reduce error
- Drive toward target

Integral

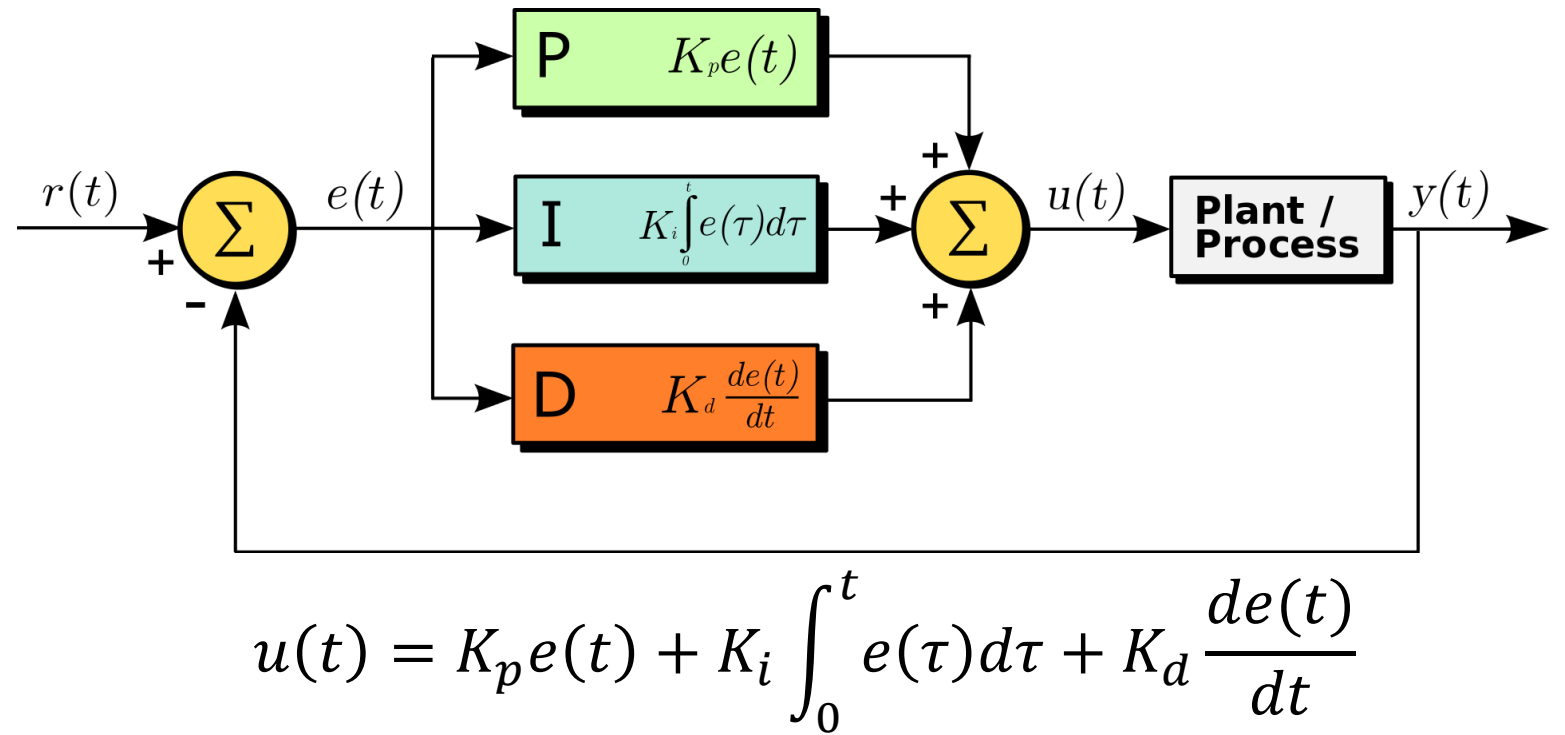
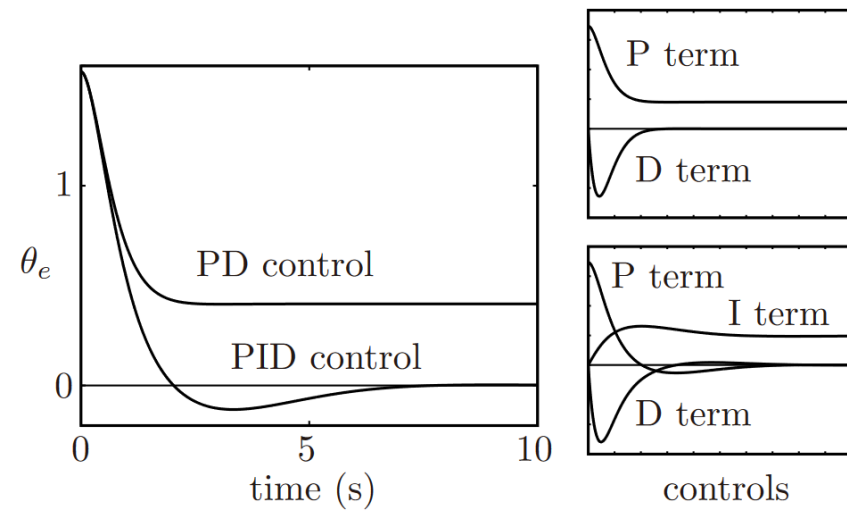
- Reduce steady-state offset
- Accumulates over time

Derivative

- Prevent overshoot
- Dampens approach



$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$



Summary:

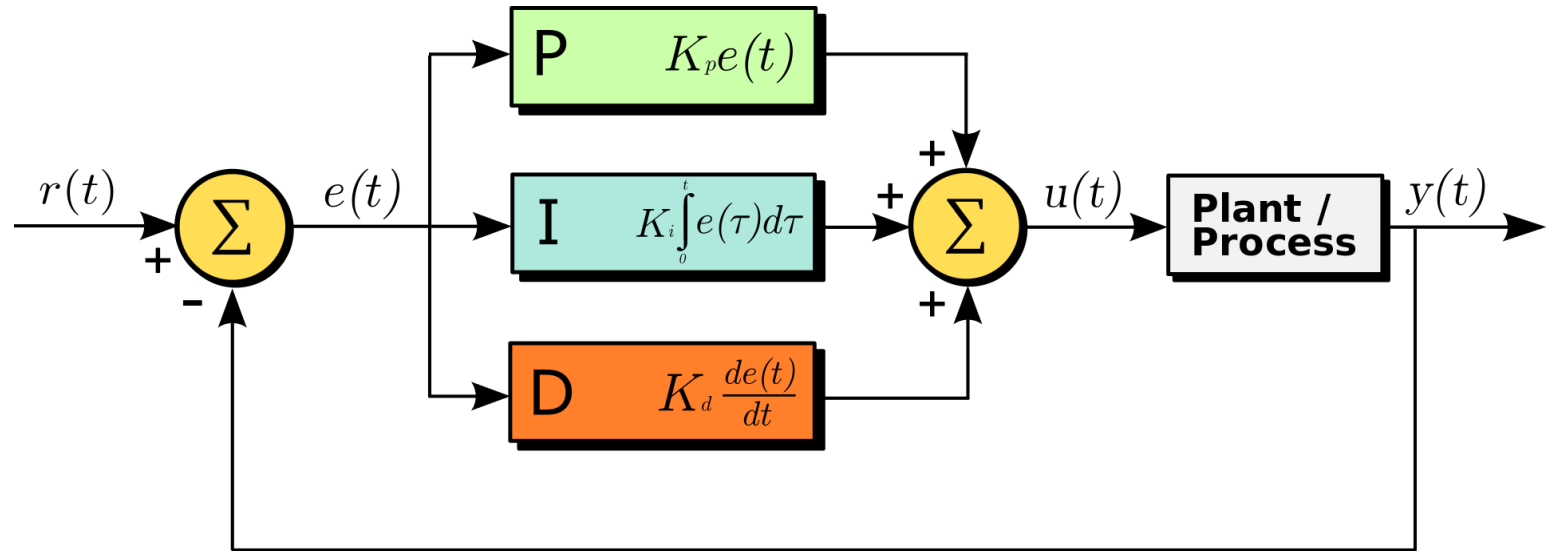
P: Fast response, but steady-state error may remain.

PI: Removes steady-state error, but can oscillate.

PD: Improves stability and speed, but still leaves steady-state error.

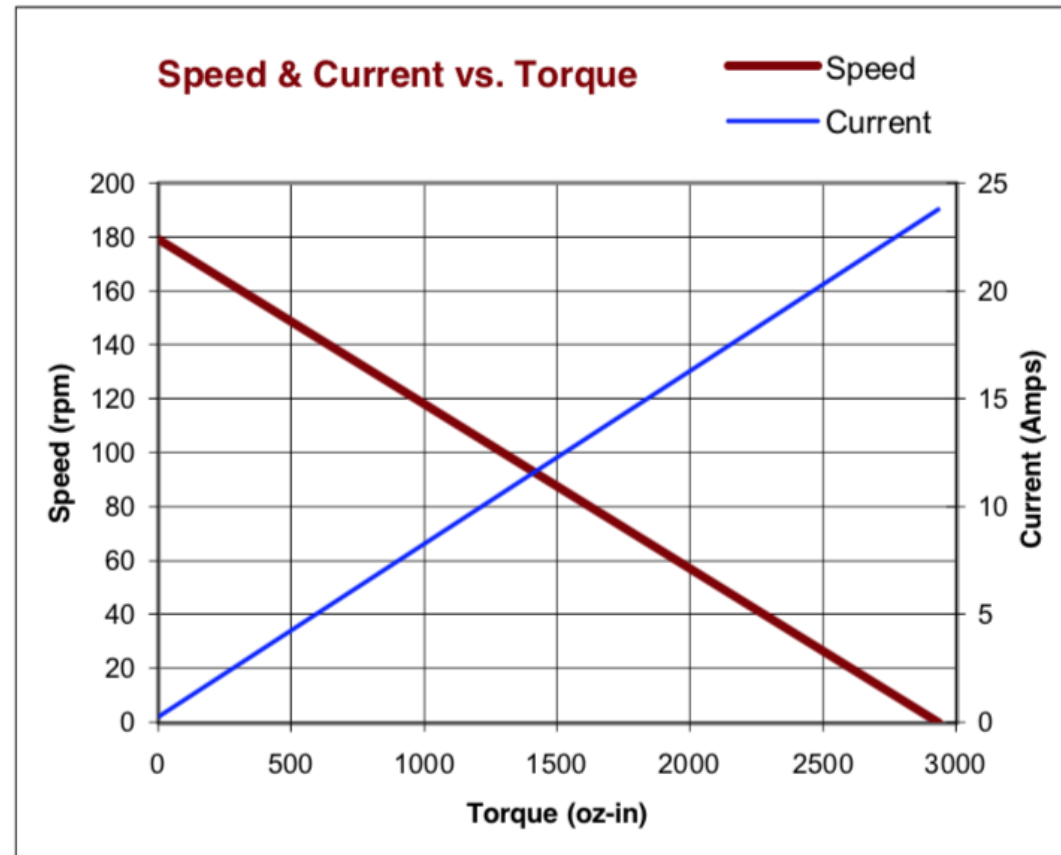
PID: Combines all three → fast, stable, and zero steady-state error.

What exactly
are we
controlling?



$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

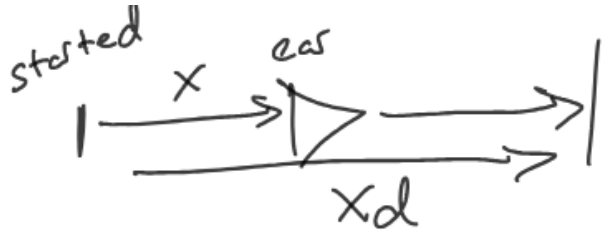
Motor speed & power vs torque



Direct Control of Velocity

- What if our controller directly controlled the velocity of our motor?
- We are ignoring the inertia of the system
- It is true for stepper motors (within reason)
- A useful approximation for DC motors

Set Point or Achievement Error



Set point error: $e = X_d - X$

X_d is a constant

Go 40 cm

p controller $u = v = \frac{dx}{dt} = K_p e$

Derivative of e : $\dot{e} = \dot{X}_d - \dot{x}$
 $\dot{X}_d = 0$
 $\dot{e} = -\dot{x}$
 $\dot{x} = K_p e$
 $\dot{e} = -K_p e$
 Diff eq.
 $\leadsto$ solution
 $e(t) = \exp(-K_p t)$

